

Module – managing SELinux

1. Check the current SELinux status on your Linux system using the `getenforce` and `sestatus` commands, and write down what each output means.

ANS :

- ❖ Command :
 - `getenforce`
 - ❖ Sample Output
 - Enforcing
 - ❖ Meaning
 - Enforcing → SELinux security policies are active and enforced.
 - Permissive → SELinux logs violations but does not block actions.
 - Disabled → SELinux is completely turned off.
2. Switch SELinux to permissive mode temporarily using the `setenforce` command, then try creating a new file in `/var/www/html` and note any differences in error messages or permissions compared to enforcing mode.

Hint: Use `setenforce 0` for permissive and `setenforce 1` to return to enforcing.

ANS :

- ❖ Check Current Mode
 - `getenforce`
 - ❖ Output:
 - Enforcing
 - ❖ Change to Permissive Mode
 - `sudo setenforce 0`
 - ❖ Verify
 - `getenforce`
 - ❖ Output:
 - Permissive
- In Enforcing Mode, SELinux may block unauthorized actions.
 - In Permissive Mode, actions are allowed but logged in audit logs.
 - No SELinux denial error is shown while in permissive mode.

3. List all files in /var/www/html and use the ls -Z command to display their SELinux security contexts. Change the context of one file to httpd_sys_content_t using chcon and verify the change.
Hint: Use chcon --type=httpd_sys_content_t filename

ANS:

- ❖ List Files with SELinux Context
 - ls -Z /var/www/html
 - ❖ Sample Output
 - -rw-r--r--. root root system_u:object_r:default_t:s0 test.html
 - ❖ Meaning
 - user:role:type:level
 - ❖ Change Context
 - sudo chcon --type=httpd_sys_content_t test.html
 - ❖ Verify
 - ls -Z /var/www/html
 - ❖ Output:
 - -rw-r--r--. root root system_u:object_r:httpd_sys_content_t:s0 test.html
4. A user is unable to upload images to a folder for their food review blog (like Zomato) hosted in /srv/foodie/images. Use the audit2why and audit2allow tools to analyze SELinux denial logs and suggest a solution to allow uploads.
Hint: Check /var/log/audit/audit.log for recent denials.

ANS:

- ❖ Assume uploads are failing in:
 - /srv/foodie/images
- ❖ **View Recent SELinux Denials**
 - sudo cat /var/log/audit/audit.log | grep denied
- ❖ **Analyze Why It Was Denied**
 - sudo cat /var/log/audit/audit.log | audit2why
- ❖ **Example Output**
 - type=AVC denied { write } for pid=1234 comm="httpd"
- ❖ Meaning:
 - Apache (httpd) is trying to write to the folder.
 - SELinux policy is blocking the write operation.